Epidemiology
Week 3
Cohort Study

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25 October 2026

1 Definition and Method

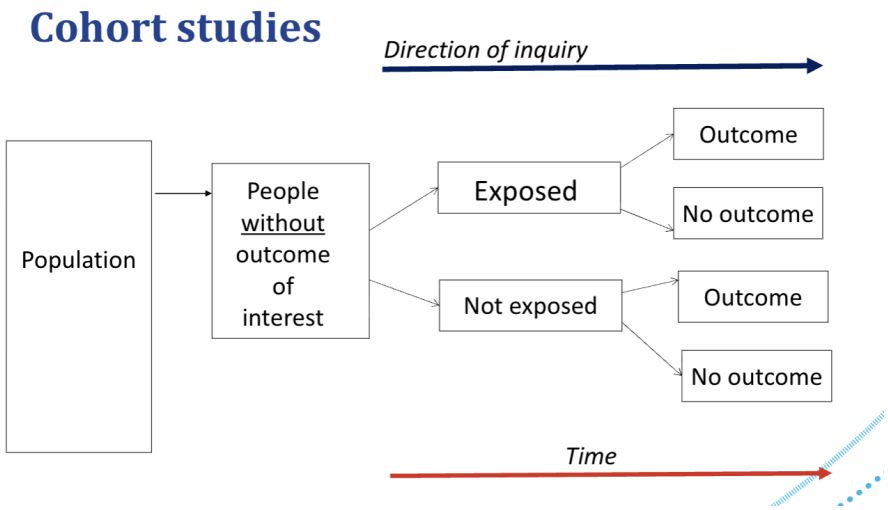


Figure 1: Cohort study design

As the image above says, starting from choosing people **WITHOUT** the outcome of interest the study population then follow-up splitting them into either the exposed or not exposed group and ending up with the outcome or no outcome from both groups.
Conducting a cohort study follows these steps:

1. Research question - Think about how exposure is associated with the outcome(s)

2. Identify the study population - This depends on how common the exposure of interest is
3. Measure the exposure
4. Follow-up participants and determine their outcome status
5. Data analysis
6. Interpret the results
7. Assess possible sources of error

Measure the exposure

Measuring the exposure can be gathered from a number of sources:

- Medical records / tests
- Measures of blood
- Physical measurements, e.g. height, weight, etc.
- Direct questioning to the subject of interest, questionnaires
- Environmental measures, e.g. water sampling

It is highly important to define the exposure, such as *how has epxosure been ascertained?* or *how often has exposure been measured?* to avoid boundary (a maybe situation) cases. Confounding variables also need to be measured.

Follow up participants and determine their outcome status

Just like defining the exposure, outcome also needs to be defined clearly and its ascertainment for the same reason to avoid boundary cases. Furthermore, things like *do all the population have similar risk of outcome?* or whether the population at risk has changed over time needs to be thought out thoroughly.

Data analysis

Cohort study calculates **incidence** as a risk or rate, which can then be compared to get the risk / rate ratio. The difference in calculating risk or rate is typically on whether the follow-up time is identical (or nearly) for the cohort members, risk is for when the follow-up time is identical, while rate is when there's variation between the follow-up time of the cohort members.

Analysis should also allow for the changing nature of exposure and for the differences in groups (by using standardisation, multivariate adjustments, Poisson regression, or Cox regression). One thing to note is **Effect modification**, it is when the effect of an exposure or treatment of an outcome changes depending on the level of a third variable, i.e. *A third variable distorts the exposure, outcome, or both*

Calculating risk

When the follow-up time is **identical** (or nearly) identical for all cohort members.

	outcome		
exposure	Yes	No	Total
exposed	a	b	a+b
not exposed	c	d	c+d

Figure 2: Calculate risk

The formulas for the risks are:

Risk in exposed, $r_e = \frac{a}{a+b}$	**Risk in unexposed**, $r_0 = \frac{c}{c+d}$
Risk ratio $= \frac{r_e}{r_0}$	**Risk difference** $= r_e - r_0$

Table 1: Calculating risk in cohort studies

Calculating rate

When there is **variation** between cohort members in the duration of follow-up time.

	outcome		
exposure	Yes	No	person-time
exposed	a	b	y_e
not exposed	c	d	y_0

Figure 3: Calculate rate

The formulas for the rates are:

Rate in exposed, $r_e = \frac{a}{y_e}$	**Risk in unexposed**, $r_0 = \frac{c}{y_0}$
Risk ratio $= \frac{r_e}{r_0}$	**Risk difference** $= r_e - r_0$

Table 2: Calculating rate in cohort studies

Interpret results

Interpreting the results follows these 3 principles:

- RR = 1 -> No association
- RR > 1 -> Risk in exposed is greater than risk in not exposed
- RR < 1 -> Risk in exposed is less than risk in not exposed

Where RR is the risk / rate ratio

Assess possible sources of error

Errors in studies can come from multiple sources, but primarily they may come from bias, confounding variables, or even chance. Examples of bias are information bias or attrition bias that came due to cohort members not available for the follow-up. It is highly important to collect information about possible confounders.

Pros and cons of cohort study

Advantage	**Disadvantage**
Temporal sequence of events can be established	Slow and usually expensive
Rare exposures can be studied	Inefficient for **rare disease**
Can examine multiple outcomes	Differential loss to follow up can lead to attrition / information bias
Can provide information about the time course of outcome (natural history)	Exposure status can change
Can measure incidence rate directly	Can become historical

Table 3: Pros and cons of cohort study

2 Types of Cohort Studies

Cohort studies are divided into three based on how the participants are followed in time.

- **Prospective** (Concurrent)
- **Retrospective** (Historical)
- **Ambidirectional**

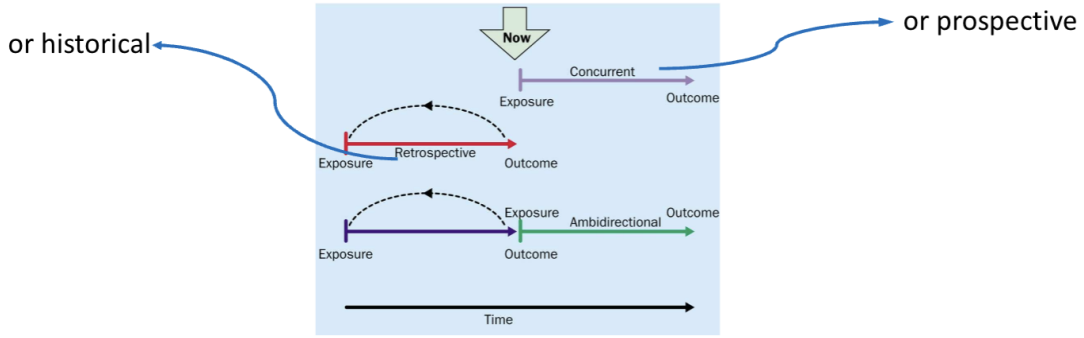


Figure 4: Cohort study types

2.1 Prospective cohort study

This study follows the cohort members forward in time. The pros and cons are

Advantage	**Disadvantage**
Temporal sequence of events can be established	Slow and usually expensive
Rare exposures can be studied	Inefficient for **rare disease**
Can examine multiple outcomes	Differential loss to follow up can lead to attrition / information bias
Can provide information about the time course of outcome (natural history)	Exposure status can change
Can measure incidence rate directly	Can become historical
Investigators define exposure and outcome	

Table 4: Pros and cons of prospective cohort study

2.2 Retrospective cohort study

This study reconstructs past exposure history and track subsequent health outcome using existing records. The pros and cons are

Advantage	**Disadvantage**
Rare exposures can be studied	Inefficient for **rare disease**
Can examine multiple outcomes	Differential loss to follow up can lead to attrition / information bias
Can provide information about the time course of outcome (natural history)	Exposure status can change
Can measure incidence rate directly	Can become historical
Temporal sequence of events can be established	**Investigator has no control over how data was collected**
Efficient, take less time to conduct and thus less expensive than prospective cohort	**Might lack information about potential confounders**
Useful for **occupational exposures**	**Can be difficult to identify appropriate exposed and not exposed groups**

Table 5: Pros and cons of retrospective cohort study

2.3 Ambidirectional cohort study

This study combines both elements from prospective and retrospective cohort study. It works by looking at past records to identify the cohort, determining their past exposures then follow them into the future to determine their health outcome.

3 Exercise on calculating risk and rates in cohort studies

3.1 Example 1

A cohort study began in 1990 and followed participants to determine whether poor sleep quality was associated with risk of incident stroke. At study recruitment, 9936 participants agreed to take part and completed the Pittsburgh Sleep Quality Index. 8040 people had good sleep quality. After 10 years of follow-up, 145 strokes had occurred in people with good sleep quality and 87 strokes had occurred in people with poor sleep quality.

	Outcome		
Exposure	**Stroke**	**No Stroke**	**Total**
Exposed - Poor Sleep	87	1809	1896
Not exposed - Good Sleep	145	7895	8040

Table 6: Example 1

Since the follow-up is on the same time, the risk is calculated here instead of the rate.

- Risk in exposed group, $r_e = \dfrac{87}{1896} = 0.046$
- Risk in not exposed group, $r_0 = \dfrac{145}{8040} = 0.018$
- Risk ratio $= \dfrac{0.046}{0.018} = 2.56$

In conclusion, people with poor sleep quality had 2.56 times the risk of stroke than people with good sleep quality.

3.2 Example 2

During 1995-2014, a cohort study followed 1,879,227 women aged 15 to 49 living in Denmark to investigate the relationship between use of hormonal contraceptives and risk of ovarian cancer. Women were followed until either: first ovarian cancer, death, emigration, age 50 or end of follow-up. The total follow-up time in the study was 21,494,781 person-years, of which 13,344,531 person-years occurred in users of hormonal contraceptives. 1249 ovarian cancers were registered, most of which were in women who did not use hormonal contraceptives (n=771 cancers).

	Outcome	
Exposure	**Ovarian Cancer**	**Person-years**
Exposed - HC user	478	13,344,531
Not exposed - HC non-user	771	8,150,250
Total	1249	21,494,781

Table 7: Example 2

The rate is calculated here since the follow-up time varies between the cohort members.

- Rate in exposed group, $r_e = \frac{478}{1,334,4531} = 0.000035824$ or 3.58 per 100,000 person-years
- Rate in non-exposed group, $r_0 = \frac{771}{8,150,250} = 0.0000946$ or 9.46 per 100,000 person-years
- Rate ratio $= \frac{\text{3.58 per 100,000 person-years}}{\text{9.46 per 100,000 person-years}} \approx 0.38$

In conclusion, women who used hormonal contraceptives had 0.38 times the rate of ovarian cancer compared to non-users.

3.3 More on Person-time

For an individual, person-time accounts for the total time period during which an individual could develop or is at risk of the outcome of interest.

For a population / cohort study, total person-time at risk is the sum of all individuals person-time, the total time during which outcomes could occur and would be considered events in the population of interest.

3.3.1 Example - Person-time

A cohort study recruits from January 1995 and follows members until lung cancer or death from another cause or end of study in December 2014.

Exposure status	**Date of entry**	**Event**	**Event date**
Smoker	Jan 1995	Lung cancer	Mar 2006
Non-Smoker	Feb 1995	Death	June 2014
Non-Smoker	Apr 1995	End of study	Dec 2014
Non-Smoker	Jan 1995	Lung cancer	Aug 2000
Smoker	Feb 1995	Lung cancer	Jul 2002
Smoker	Jun 1995	End of study	Dec 2014
Smoker	Mar 1995	Death	Nov 2001
Non-Smoker	Jan 1995	End of study	Dec 2014
Non-Smoker	Mar 1995	End of study	Dec 2014
Smoker	Jan 1995	Lung cancer	Feb 2012

Table 8: Person-time example

Calculate the person-years and the rates.

Exposure status	**Date of entry**	**Event**	**Event date**	**Person-years**
Smoker	Jan 1995	Lung cancer	Mar 2006	11.25
Non-Smoker	Feb 1995	Death	June 2014	19.42
Non-Smoker	Apr 1995	End of study	Dec 2014	19.75
Non-Smoker	Jan 1995	Lung cancer	Aug 2000	5.67
Smoker	Feb 1995	Lung cancer	Jul 2002	7.50
Smoker	Jun 1995	End of study	Dec 2014	19.58
Smoker	Mar 1995	Death	Nov 2001	6.75
Non-Smoker	Jan 1995	End of study	Dec 2014	20.00
Non-Smoker	Mar 1995	End of study	Dec 2014	19.83
Smoker	Jan 1995	Lung cancer	Feb 2012	17.17

Table 9: Person-time example calculated

The first row gives 11.25 person-years because you just take the difference between March 2006 and January 1995 which would be 11 years and 3 months or 11.25. The rates are as follows,

- Rate in exposed, $r_e = \dfrac{\text{Total smokers with lung cancer}}{\text{Total person-years from all smokers}}$

$$= \frac{3}{11.25 + 7.50 + 19.58 + 6.75 + 17.17}$$

$$\approx 0.0482 \text{ or } 4.82 \text{ per } 100 \text{ person-years}$$

- Rate in non-exposed, $r_0 = \dfrac{\text{Total non-smokers with lung cancer}}{\text{Total person-years from all non-smokers}}$

$$= \frac{1}{19.42 + 19.75 + 5.67 + 20.00 + 19.83}$$

$$\approx 0.0118 \text{ or } 1.18 \text{ per } 100 \text{ person}$$

- Rate ratio $= \dfrac{4.82 \text{ per } 100 \text{ person-years}}{1.18 \text{ per } 100 \text{ person-years}} = 4.08$